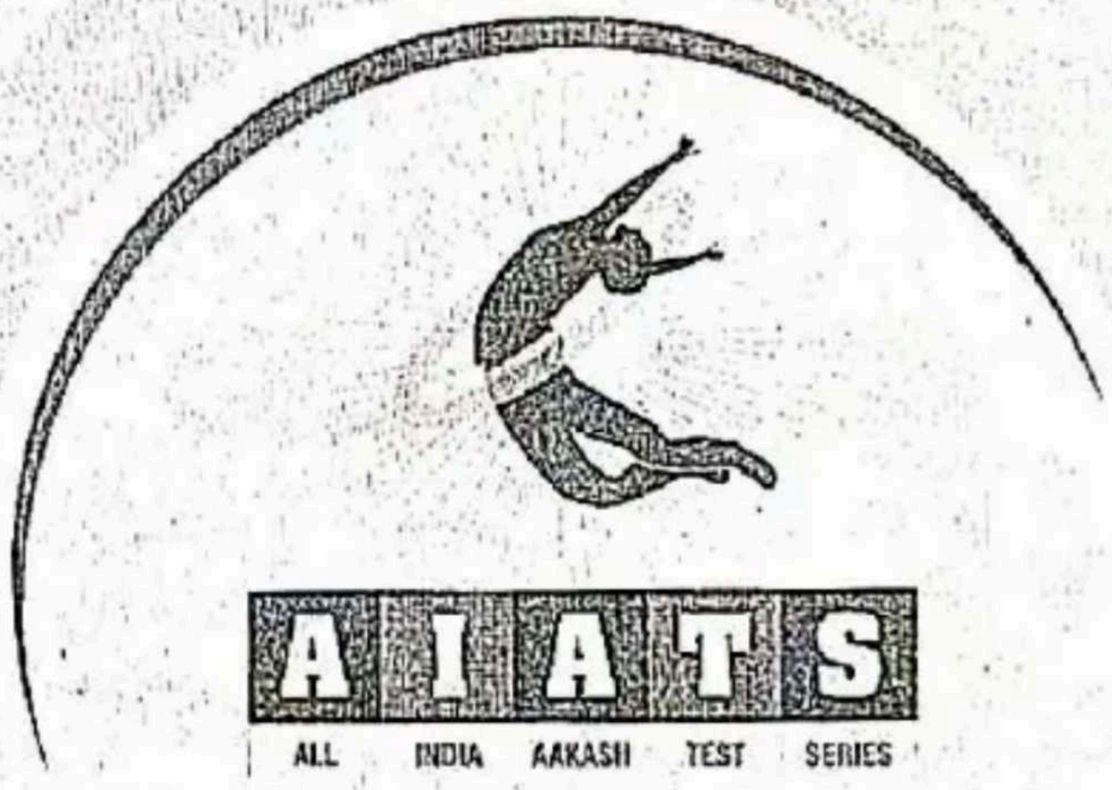
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Test Date: 04-12-2022

Aakash BYJU'S

E
Booklet Code



for

Medical Entrance Exam - 2023

National Eligibility-cum-Entrance Test (NEET)

TEST No. 1

(XII Passed Students)

INSTRUCTIONS FOR CANDIDATES

1. Read each question carefully.
2. It is mandatory to use Blue/Black Ball Point Pen to darken the appropriate circle in the answer sheet.
3. Mark should be dark and should completely fill the circle.
4. Rough work must not be done on the answer sheet.
5. Do not use white-fluid or any other rubbing material on answer sheet. No change in the answer once marked is allowed.
6. Student cannot use log tables and calculators or any other material in the examination hall.
7. Before attempting the question paper, student should ensure that the test paper contains all pages and no page is missing.
8. Each correct answer carries four marks. One mark will be deducted for each incorrect answer from the total score.
9. Before handing over the answer sheet to the invigilator, candidate should check that Roll No. and Centre Code have been filled and marked correctly.
10. Immediately after the prescribed examination time is over, the answer sheet to be returned to the invigilator.
11. There are two sections in each subject i.e., Section-A & Section-B. You have to attempt all 35 questions from Section-A & only 10 questions out of 15 from Section-B.

Note : It is compulsory to fill **Roll No.** and **Test Booklet Code** on answer sheet, otherwise your answer sheet will not be considered.

Test No. 1

TOPICS OF THE TEST

Physics

Physical World, Units and Measurements, Motion in a Straight Line, Motion in a Plane

Chemistry

Some Basic Concepts of Chemistry, Structure of Atom

Botany

Cell: The Unit of Life, Cell Cycle and Cell Division, The Living World

Zoology

Structural Organisation in Animals – Animal Tissues, Biomolecules

MM : 720

TEST - I

Time : 3 Hrs. 20 Min.

[PHYSICS]

Choose the correct answer:

SECTION-A

1. Parsec is the unit of
   (1) Time (2) Angle
   (3) Distance (4) Mass

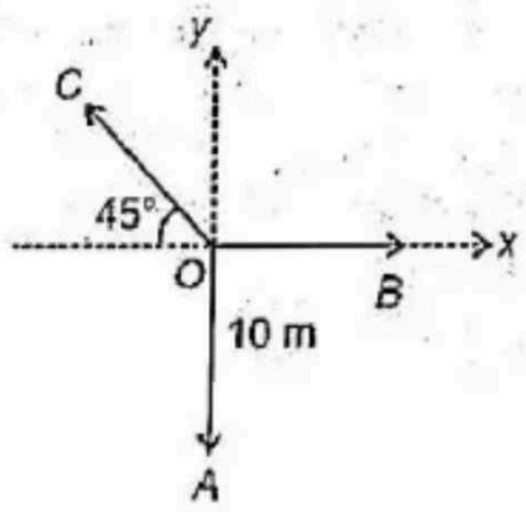
2. Each side of a cube is measured to be 2.32 m. The total surface area of the cube with appropriate significant figure is
   (1) 32.2944 m^2 (2) 32.3 m^2
   (3) 32.29 m^2 (4) 32.294 m^2

3. Column I contains the four fundamental forces and column II contains their ranges. Match the column I with column II.

Column I	Column II
A. Gravitational force	P. 10^{-16} m
B. Electromagnetic force	Q. 10^{-15} m
C. Strong nuclear force	R. Infinite
D. Weak nuclear force	

   (1) A → P, B → R, C → Q, D → P
   (2) A → R, B → R, C → Q, D → P
   (3) A → R, B → P, C → Q, D → R
   (4) A → Q, B → P, C → P, D → Q

4. The sum of the three vectors shown in the figure is zero. The magnitudes of the vectors $\overrightarrow{OB}$ and $\overrightarrow{OC}$ are respectively, If $|\overrightarrow{OA}| = 10$ m



   (1) 10 m, $10\sqrt{2}$ m (2) $10\sqrt{2}$ m, $10\sqrt{2}$ m
   (3) $10\sqrt{2}$ m, 20 m (4) 20 m, $10\sqrt{2}$ m

5. A physical quantity X is defined by $X = \frac{A^2\sqrt{B}}{C^2}$, and the percentage errors in the measurements of A, B and C are 1%, 2% and 0.5% respectively.
   The maximum percentage error in calculation of X is
   (1) 2% (2) 4%
   (3) 3% (4) 6%

6. The least count of a watch is 0.01 s. The time of 40 oscillations of a pendulum is measured to be 40 seconds. The maximum percentage error in the measurement of time will be
   (1) 25% (2) 4%
   (3) 0.25% (4) 0.025%

7. A unitless quantity
   (1) May not be dimensionless
   (2) Is always dimensionless
   (3) Is not always dimensionless
   (4) Does not exist

8. The SI unit of mass is now defined in terms of
   (1) Planck's constant
   (2) Boltzmann constant
   (3) Gravitational constant
   (4) Stephan's constant

9. Which among the following is dimensionally incorrect?
   (Here h → height, T → surface tension, ρ → density, P → Pressure, V → volume, η → coefficient of viscosity, ν → frequency, I → moment of inertia, g → acceleration due to gravity, R → radius, t → time, m → mass, L → length)
   (1) $h = \frac{2T\cos\theta}{\rho Rg}$ (2) $\nu = \sqrt{\frac{P}{\rho}}$
   (3) $V = \frac{P \cdot \pi R^4 t}{8\eta L}$ (4) $\nu = 2\pi\sqrt{\frac{I^2}{mgL}}$

@DefeatNEET

10. If dimensions of a physical quantity are given by $M^aL^bT^c$, then the physical quantity will be
(1) Torque if $a = 2, b = 1, c = -2$
(2) Force if $a = 1, b = 2, c = -1$
(3) Strain if $a = 0, b = 0, c = +1$
(4) Stress if $a = 1, b = -1, c = -2$

11. In summers, the daytime temperature is recorded as $(46 \pm 1)^\circ C$ and in night it is recorded as $(38 \pm 1)^\circ C$.
The fall in temperature should be reported as
(1) $(8 \pm 0)^\circ C$ (2) $(8 \pm 1)^\circ C$
(3) $(8 \pm 2)^\circ C$ (4) $(78 \pm 2)^\circ C$

12. Consider the following statements.
a. 00.020480 has two significant zeros
b. 0.475 is equal to 0.48 in two significant figures
Choose the incorrect statement(s).
(1) a only (2) b only
(3) Both a and b (4) Neither a nor b

13. If force (F), velocity (v) and time (T) are taken as the fundamental units, then the representation of Young's modulus 'Y' will be
(1) $[F^0vT^{-2}]$ (2) $[Fv^{-2}T^{-2}]$
(3) $[F^2v^{-1}T^{-1}]$ (4) $[F^2vT^2]$

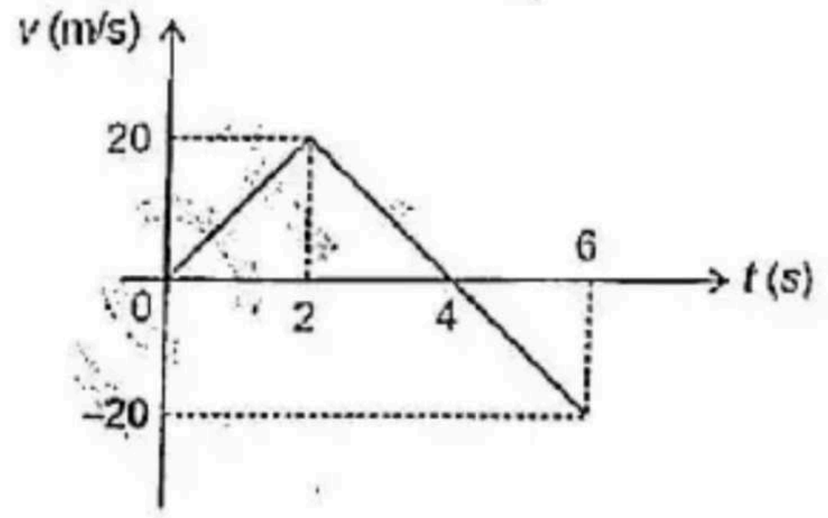
14. Which among the following is/are dimensionless?
(1) Poisson's ratio (2) Relative density
(3) Density (4) Both (1) and (2)

15. Which among the following can be deduced using dimensional method?
(1) $4mv^2 + mgh$ (2) $a\sin\omega t + b\cos\omega t$
(3) $v = u + at$ (4) $v = at$

16. A person goes to hospital, for taking booster dose of the vaccination of Covid-19 from his home, with uniform speed 20 m/s and stays in hospital for $\frac{1}{2}$ hour and then returns to home with uniform speed of 10 m/s. If the distance between home and hospital is 500 m then average speed and average velocity for complete journey will be respectively
(1) 0.53 m/s, zero
(2) Zero, 0.33 m/s
(3) 0.83 m/s, 0.33 m/s
(4) 0.53 m/s, 13.33 m/s

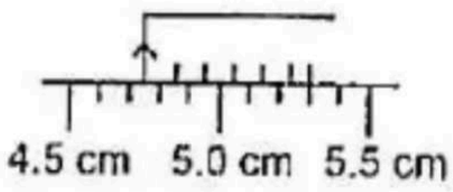
17. A particle is moving along a straight line with uniform acceleration. If position (x) is given in terms of time (t) as $x = (5t^2 + 4t + 8)$ m. Then
(1) Its initial velocity is 18 m/s
(2) Its velocity at time $t = 3$ s is 86 m/s
(3) Its acceleration is 5 m/s^2
(4) Position at $t = 2$ s is $x = 36$ m

18. The velocity (v) - time (t) graph for a particle moving along x axis is shown in the figure. The average velocity in the time interval $t = 0$ to $t = 6$ s is



(1) $\frac{5}{3}$ m/s (2) $\frac{10}{3}$ m/s
(3) 5 m/s (4) 2 m/s

19. The diagram shows the part of vernier scale on a pair of calipers. The correct reading is (least count = 0.01 cm)



(1) 4.74 cm (2) 4.56 cm
(3) 4.70 cm (4) 4.76 cm

20. The angular velocity of a particle moving in circle of radius 50 cm, is increased from 200 revolution per minute to 440 revolution per minute in 2 minutes. The tangential acceleration of the particle is
(1) $\frac{\pi}{30}$ m/s^2
(2) $\frac{\pi}{15}$ m/s^2
(3) $\frac{1}{30}$ m/s^2
(4) 30 m/s^2

21. The horizontal range and maximum height of a projectile is R and H respectively. If a constant horizontal acceleration $\frac{g}{4}$ is imparted to the projectile due to wind, then its new increased horizontal range will be (g is acceleration due to gravity)

(1) $2R$ (2) $R+H$

(3) $R+2H$ (4) $R+\frac{H}{4}$

22. A particle moving in x - z plane have x and z co-ordinate at any time t as $x = 2 + 7t + 4t^2$ and $z = 5t$, where x and z are in metre and t in second. The acceleration of the particle at time $t = 2$ s is

(1) $\sqrt{89}$ m/s^2 (2) $\sqrt{69}$ m/s^2

(3) 8 m/s^2 (4) 13 m/s^2

23. A particle is moving is x - y plane with velocity $\vec{v} = (2-t)\hat{i} + \frac{t^2}{8}\hat{j}$, where t is in s and v in m/s. If the particle starts from origin, then y-coordinate of particle when its x-coordinate becomes zero after start, will be

(1) (16/3) m (2) (7/3) m

(3) (14/3) m (4) (8/3) m

24. A man can swim at speed of 3 km/h in still water. He wants to cross a 500 m wide river flowing at 2 km/h. The man keeps himself always at an angle 127° with the river flow while swimming. The point on the opposite bank, where he will arrive is

(1) $\frac{125}{3}$ m in the direction of river flow from opposite point on the other bank

(2) $\frac{250}{3}$ m in the direction of river flow from opposite point on the other bank

(3) $\frac{125}{3}$ m in opposite direction of river flow from the opposite point on the other bank

(4) $\frac{250}{3}$ m in opposite direction of river flow from the opposite point on the other bank

25. A particle A is moving in x - y plane with velocity $(3\hat{i} + 4\hat{j})$ m/s and particle B is moving in y - z plane with velocity $(3\hat{j} + 4\hat{k})$ m/s. The magnitude of velocity of particle B w.r.t. velocity of particle A is closest to

(1) Zero (2) 5 m/s

(3) 10 m/s (4) 20 m/s

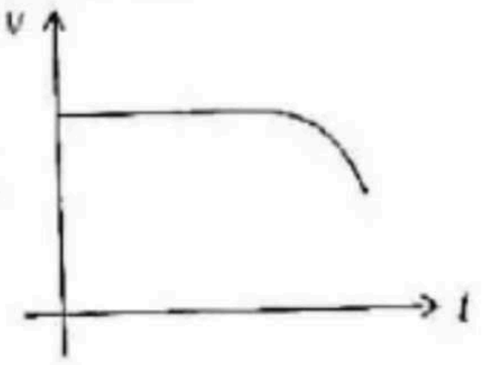
26. The equation of trajectory of projectile is $y = \sqrt{3}x - 10x^2$. The angle of projection and initial speed of projection are respectively ($g = 10$ m/s^2).

(1) 30°, $\sqrt{2}$ m/s (2) 60°, $\sqrt{2}$ m/s

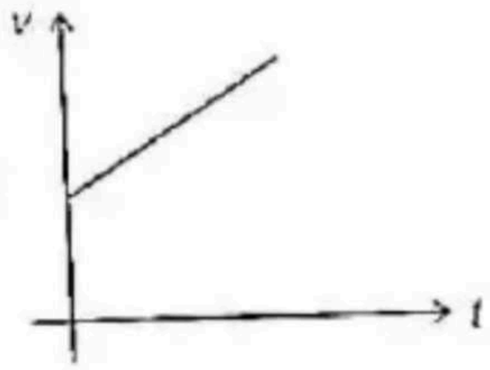
(3) 60°, 2 m/s (4) 30°, 1 m/s

27. Which of the following represents the motion of a particle in straight line, with uniform (non-zero) acceleration?

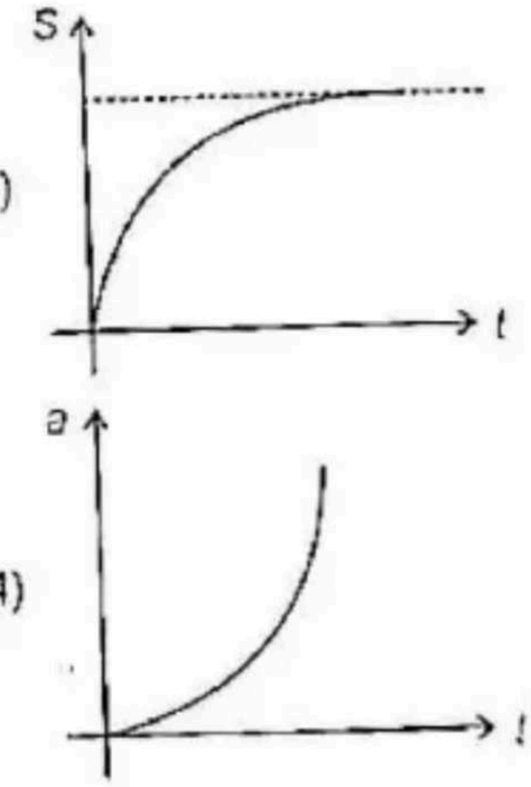
(1)



(2)



(3)



(4)

28. A particle located at origin at time $t = 0$, starts moving along positive y direction with a velocity v that varies as $v = \sqrt{y}$. The displacement of the particle varies with time as

(1) $\frac{t^3}{3}$ (2) $\frac{t^2}{2}$

(3) $\frac{t^2}{4}$ (4) $2t$

29. If relation between time (t) and position (x) is $t = 3x^2 + 2x$. Then relation between velocity and acceleration is

(1) $a \propto -v^{3/2}$ (2) $a \propto -v^3$

(3) $a \propto -v^2$ (4) $a \propto -\frac{1}{v^2}$

30. At time t (in second), positions of three particles A, B and C are as follows

$x_A = (2t + 8)$ m

$x_B = (3t^2 + 2t + 14)$ m

$x_C = (5t^3 + 4t^2 + 8t)$ m

Then we can conclude that

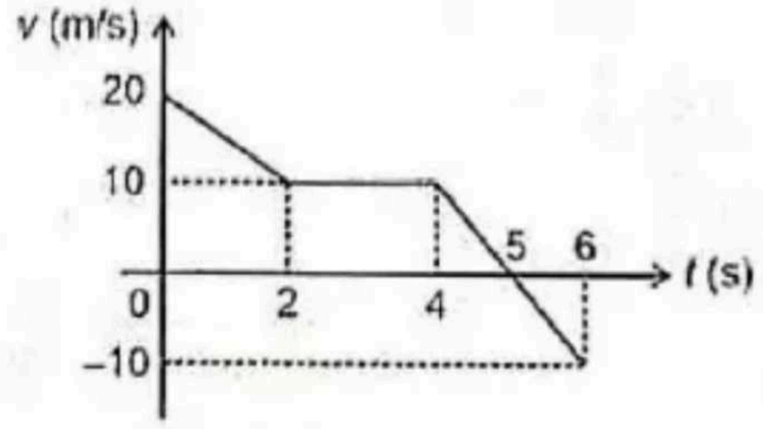
(1) All the particles are moving with constant non-zero acceleration

(2) Particle A moves with constant non-zero acceleration

(3) Particle B moves with constant non-zero acceleration

(4) Particle C moves with constant non-zero acceleration

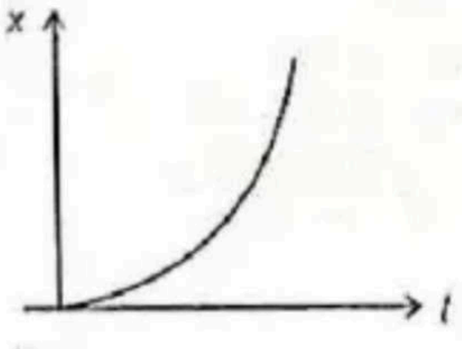
31. In the given velocity-time (v - t) graph the distance travelled by a particle in 6 s will be

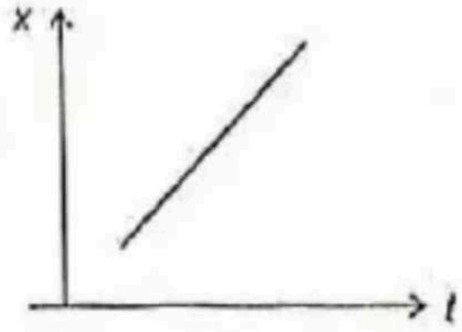


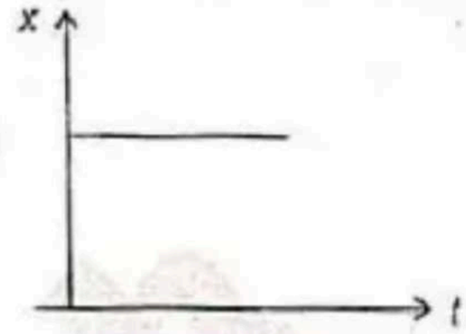
(1) 60 m (2) 50 m

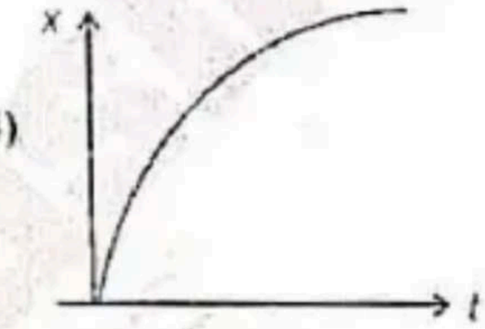
(3) 40 m (4) 30 m

32. The position-time graph for the motion with negative acceleration is

(1)


(2)


(3)


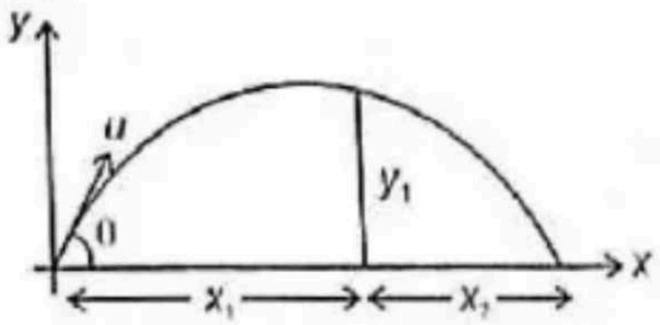
(4)


33. A particle moves in x - y plane according to the rule, $x = a\sin\omega t$ and $y = a\cos\left(\omega t + \frac{\pi}{2}\right)$. The particle follows

(1) An elliptical path (2) A circular path

(3) A parabolic path (4) A straight line path

34. A projectile is projected from the ground as shown in the figure.



The correct relation among the x_1, x_2, y_1 and θ is

(1) $y_1 = \left(\frac{x_1x_2}{x_1 + x_2}\right)\tan\theta$ (2) $y_1 = \left(\frac{x_1x_2}{x_1 - x_2}\right)\tan\theta$

(3) $y_1 = \left(\frac{x_1x_2}{x_1 + x_2}\right)\sec^2\theta$ (4) $y_1 = \left(\frac{x_1x_2}{x_1 - x_2}\right)\cos^2\theta$

35. Change in a vector may occur due to
(1) Rotation of frame of reference
(2) Translation of frame of reference
(3) Rotation of vector
(4) Both (1) and (2)

SECTION-B

36. A particle is projected from ground, such that at the maximum height of projectile, velocity of particle becomes $\frac{\sqrt{3}}{2}$ times its initial velocity (u).

From this information match the column I with column II

Column I		Column II	
A.	Square of time of flight of projectile	P.	$\frac{3u^2}{4g}$
B.	Maximum height of the projectile	Q.	$\frac{\sqrt{3}u^2}{2g}$
C.	Horizontal range of the projectile	R.	$\frac{u^2}{g^2}$
D.	Radius of curvature at maximum height	S.	$\frac{u^2}{8g}$

(1) A → R, B → P, C → Q, D → S
(2) A → R, B → Q, C → S, D → P
(3) A → P, B → R, C → Q, D → S
(4) A → R, B → S, C → Q, D → P

37. Two vectors $\vec{A}$ and $\vec{B}$ are given as $\vec{A}=3\hat{i}$ and $\vec{B}=\hat{j}$. The component of vector $\vec{A}$ in direction of vector $\vec{B}$ is
(1) $\frac{(\hat{i}-\hat{j})}{\sqrt{2}}$
(2) $\frac{(\hat{i}-\hat{j})}{2}$
(3) $\frac{(-\hat{i}+\hat{j})}{\sqrt{2}}$
(4) Zero

38. Two projectiles A and B are projected with angle of projections 25° and 45° respectively. If R_A and R_B are their respective horizontal ranges, then
(1) $R_A = R_B$ (2) $R_A > R_B$
(3) $R_A < R_B$ (4) Data is insufficient

39. Six particles situated at the corners of a regular hexagon of side 4 m, moves with a constant speed 2 m/s. Each particle maintains a direction towards the particle at the next corner. The time at which the particles will meet each other, will be
(1) 4 s (2) 2 s
(3) 1 s (4) 3 s

40. Rain is falling vertically downwards with speed 18 km/h, wind starts blowing with 2.5 m/s from east to west direction. In which direction should a boy waiting for school bus hold his umbrella?
(1) At an angle $\tan^{-1}(2)$ with vertical towards east
(2) At an angle $\tan^{-1}\left(\frac{1}{2}\right)$ with vertical towards east
(3) At angle $\tan^{-1}(2)$ with vertical towards west
(4) At an angle $\tan^{-1}\left(\frac{1}{2}\right)$ with vertical towards west

41. A particle is moving with constant speed on a circular path centred at origin. At one point (x = 2 m, y = 0), the particle has velocity $(-4\hat{j})$ m/s. The acceleration of the particle at this point is
(1) $(-8\hat{i})\ m/s^2$
(2) $(-8\hat{j})\ m/s^2$
(3) $(8\hat{i})\ m/s^2$
(4) $(-8\hat{k})\ m/s^2$

42. A particle is projected with velocity u, at an angle 53° from horizontal. If time taken by the velocity of particle to makes an angle 37° from the horizontal is 1 s, then u is nearest to
(1) 40 m/s
(2) 30 m/s
(3) 18 m/s
(4) 60 m/s

43. A class XI student measured the length of a rod and wrote it as 7.50 cm. Which instrument did he use to measure it?

(1) A meter scale

(2) A vernier calliper where the 10 divisions on vernier scale is equal to 9 divisions on main scale with 1 MSD = 1 mm

(3) A screw gauge having 50 divisions on the circular scale and pitch as 1 mm

(4) A screw gauge having 100 divisions on the circular scale and pitch as 1 mm

44. The equation of state of a gas is given by $\left(P+\frac{a}{V}\right)\left(V-\sqrt{b}\right)=cT$, where P, V and T are pressure, volume and temperature respectively and a, b and c are dimensional constants, the dimension of [ab] is

(1) $[ML^8T^{-2}]$

(2) $[ML^2T^{-2}]$

(3) $[ML^6T^{-2}]$

(4) $[M^0L^6T^0]$

45. In a new system of units called as 'Aakash unit', unit of mass is taken as 10 kg, unit of length is taken as 1 km and unit of time is 1 minute. The value of 1 J in 'Aakash unit' system is

(1) 3.6×10^3 unit (2) 3.6×10^{-5} unit

(3) 3.6×10^{-4} unit (4) 3.8×10^4 unit

46. Column I contains some name of units of length and column II contains its value in SI unit. Match the column I with column II

	Column I		Column II
A.	Astronomical unit	P	10^{-15} m
B.	Light year	Q	10^{-10} m
C.	Fermi	R	1.496×10^{11} m
D.	Angstrom	S	9.46×10^{15} m

(1) A → R, B → Q, C → P, D → S

(2) A → R, B → S, C → P, D → Q

(3) A → P, B → Q, C → R, D → S

(4) A → R, B → P, C → S, D → Q

47. A particle is dropped from rest from a large height. Assume g to be constant throughout the motion. The time taken by it to fall through successive distances of 2 m each, will be

(1) All equal, being equal to $\sqrt{\frac{4}{g}}$ s

(2) In ratio; $\sqrt{1}:\sqrt{2}:\sqrt{3}:...$

(3) In ratio; $\sqrt{1}:(\sqrt{2}-\sqrt{1}):(\sqrt{3}-\sqrt{2}):...$

(4) In ratio; $\frac{1}{\sqrt{1}}:\frac{1}{\sqrt{2}}:\frac{1}{\sqrt{3}}...$

48. A ball is projected upwards with speed 27 m/s. The distance travelled in 3rd second of motion is equal to ($g = 10$ m/s^2)

(1) 2 m (2) 2.9 m

(3) 72.9 m (4) 7.9 m

49. Two trains A and B are moving in opposite direction on parallel tracks. The length of train A is 100 m and length of train B is 80 m. The velocity of shorter train is two times to that of the longer train. If the trains take 3 s to cross each other, then the velocities of the trains are

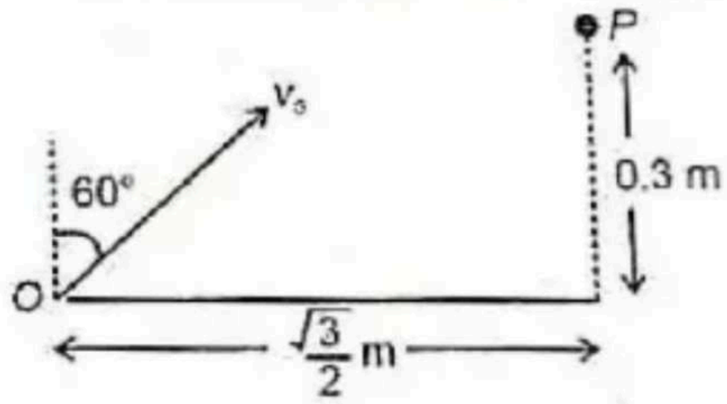
(1) $v_A = 20$ m/s, $v_B = 40$ m/s

(2) $v_A = 10$ m/s, $v_B = 20$ m/s

(3) $v_A = 20$ m/s, $v_B = 20$ m/s

(4) $v_A = 15$ m/s, $v_B = 30$ m/s

50. A ball is projected from point O with some velocity v_0 at an angle 60° with the vertical as shown in the figure. Consider a target at point 'P'. The ball will hit the target if it is projected with v_0 equal to



(1) 5 m/s

(2) 6 m/s

(3) 7 m/s

(4) 10 m/s

[CHEMISTRY]

SECTION-A

51. Number of molecules of glucose present in 400 mL of 1 M glucose solution is

(1) 3.01×10^{21} (2) 6.02×10^{23}

(3) 1.2×10^{22} (4) 2.4×10^{23}

52. If mass percentage of sulphur in a protein molecule is 0.2% then the number of sulphur atoms present in 25 g of protein is

(1) 9.4×10^{20} (2) 2.4×10^{21}

(3) 3.1×10^{19} (4) 3.6×10^{22}

53. 6.25 g of an impure sample of limestone gives 2.2 g of CO_2 on complete decomposition. The percentage of $CaCO_3$ present in the sample is

(1) 75% (2) 95%

(3) 80% (4) 85%

54. If mass ratio of methane and sulphur dioxide is 2 : 5 then the ratio of mole of their atoms will be

(1) 4 : 5 (2) 8 : 5

(3) 5 : 3 (4) 8 : 3

55. Ratio of the radius of the first orbit of Li^{2+} to the third orbit of He^{+} will be

(1) 2 : 9 (2) 2 : 27

(3) 4 : 27 (4) 1 : 8

56. Orbital angular momentum of electron present in 3*p* orbital is

(1) Zero (2) $\sqrt{6}\,h$

(3) $\sqrt{2}\,h$ (4) $2\sqrt{3}\,h$

57. If a 44 watt bulb emits monochromatic radiation of wavelength 900 nm then the number of photons emitted per second by the bulb is (Planck's constant $= 6.6 \times 10^{-34}$ Js)

(1) 4×10^{22} (2) 2×10^{20}

(3) 8×10^{24} (4) 4×10^{21}

58. An element X has two isotopes, ^{54}X and ^{56}X. If mass percentage of heavier isotope is 40% then average atomic mass of the element X will be

(1) 55.2 u (2) 54.8 u

(3) 55.4 u (4) 55.0 u

59. Volume of O_2 required at STP to react completely with 64 g of methane is

(1) 179.2 L (2) 89.6 L

(3) 134.4 L (4) 224 L

60. Correct order of energy of orbitals for filling electrons in multielectronic species is

(1) $4f > 5d > 5s > 5p$ (2) $4f > 5d > 5p > 5s$

(3) $5d > 5p > 5s > 4f$ (4) $5d > 4f > 5p > 5s$

61. Consider the following sets of quantum numbers

	n	l	m	s
a.	4	2	–1	$+\frac{1}{2}$
b.	3	1	–2	$-\frac{1}{2}$
c.	5	3	0	$+\frac{1}{2}$
d.	2	2	–1	$+\frac{1}{2}$

Which of the following sets of quantum numbers are not possible?

(1) (a) and (b) only (2) (b), (c) and (d) only

(3) (b) and (d) only (4) (a), (b) and (d) only

62. If a particle of mass 10 mg is moving with a velocity of 600 m/s then the de Broglie wavelength associated with the moving particle will be (Planck's constant (h) = 6.6×10^{-34} Js)

(1) 1.1×10^{-31} m (2) 1.1×10^{-34} m

(3) 1.1×10^{-32} m (4) 1.1×10^{-30} m

63. Maximum number of electrons accommodated in the orbitals of an atom for which n = 4 and l = 2 is

(1) 6 (2) 14

(3) 5 (4) 10

64. Number of electrons present in p orbitals of a gallium atom is (Atomic number of gallium = 31)

(1) 10

(2) 12

(3) 13

(4) 14

65. **Statement I:** Cathode rays consist of negatively charged particles, called electrons.

Statement II: The characteristics of cathode rays depend upon the nature of gas present in the cathode ray tube.

In the light of the above statements, choose the correct answer.

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

66. 2.408×10^{21} molecules of urea are present in 200 mL of aqueous solution. The molarity of the solution is

(1) 0.1 M

(2) 0.5 M

(3) 0.02 M

(4) 0.04 M

67. 21 g of N_2 is reacted with 4 g of H_2. The mass of ammonia produced after completion of the reaction is

(1) 22.61 g

(2) 25.0 g

(3) 19.25 g

(4) 17.21 g

68. **Assertion (A):** According to Heisenberg uncertainty principle, it is impossible to determine simultaneously, the exact position and exact momentum of an electron.

Reason (R): The effect of Heisenberg uncertainty principle is significant both for the motion of microscopic objects and macroscopic objects.

In the light of the above statements, choose the correct answer.

(1) (A) is correct but (R) is not correct

(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(3) Both (A) and (R) are correct but (R) is the correct explanation of (A)

(4) (A) is not correct but (R) is correct

69. Total energy and kinetic energy of electron present in second orbit of He^+ ion respectively are

(1) –3.4 eV and 3.4 eV

(2) –13.6 eV and 3.4 eV

(3) –13.6 eV and 13.6 eV

(4) –27.2 eV and 13.6 eV

70. Which among the following metals has lowest value of work function?

(1) Mg (2) Li

(3) Cu (4) K

71. Total number of nodes and radial nodes present in 5p osrbitals respectively are

(1) 5 and 3 (2) 4 and 3

(3) 3 and 2 (4) 4 and 2

72. 100 mL of 4 M NaOH solution was mixed with 400 mL of 2 M NaOH solution. The molarity of resultant solution becomes

(1) 3.2 M (2) 2.8 M

(3) 2.4 M (4) 3.0 M

73. Which among the following contains lowest number of molecules?

(1) 22.4 L of CH_4 at STP

(2) 2 g molecule of N_2

(3) 34.2 g of sugar ($C_{12}H_{22}O_{11}$)

(4) 18 mL of H_2O (density of water, 1 g/mL)

74. If mass percentage of Zn^{2+} in an enzyme is 0.327%, then the minimum molar mass of the enzyme will be (Atomic mass of Zn = 65.4 u)

(1) 4000 g mol^{-1} (2) 80000 g mol^{-1}

(3) 20000 g mol^{-1} (4) 10000 g mol^{-1}

75. In hydrogen atom, the de Broglie wavelength of the electron in the third Bohr's orbit is (Bohr's radius, a_0 = 52.9 pm)

(1) 105.8π pm (2) 317.4π pm

(3) 211.6π pm (4) 423.2π pm

76. The number of protons, neutrons and electrons in $^{88}_{38}Sr$ respectively are

(1) 38, 50 and 50

(2) 88, 38 and 50

(3) 38, 50 and 38

(4) 38, 126 and 50

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77. If mole fraction of ethanol in water is 0.18, then molality of ethanol in the solution will be

(1) 6.5 m (2) 8.2 m

(3) 10.4 m (4) 12.2 m

78. **Statement I:** Zeros preceding to first non-zero digit are significant.

Statement II: 0.08 has three significant figures.

In light of the above statements, choose the correct answer.

(1) Statement I is correct but statement II is incorrect

(2) Statement I is incorrect but statement II is correct

(3) Both statement I and statement II are incorrect

(4) Both statement I and statement II are correct

79. Which of the following series of transitions in the spectrum of hydrogen atom does not fall in infrared region?

(1) Pfund (2) Lyman

(3) Brackett (4) Paschen

80. Correct order of energy of 2*s* orbital of potassium, lithium and sodium is

(1) $E_{2s}(K) > E_{2s}(Na) > E_{2s}(Li)$

(2) $E_{2s}(Na) > E_{2s}(K) > E_{2s}(Li)$

(3) $E_{2s}(Li) > E_{2s}(Na) > E_{2s}(K)$

(4) $E_{2s}(Li) > E_{2s}(K) > E_{2s}(Na)$

81. The maximum number of orbitals that can be identified by the following quantum numbers in an atom is

$n = 5, l = 2, m = -1$

(1) 4 (2) 3

(3) 2 (4) 1

82. Suppose the elements X and Y combine to form two compounds XY and XY_3. If 0.2 mole of XY weigh 24 g and 0.1 mole of XY_3 weigh 16 g then atomic weights of X and Y respectively are

(1) 80 and 160 (2) 80 and 20

(3) 60 and 60 (4) 100 and 20

83. If the density of methanol is 0.9 g/mL, then volume of methanol needed for making 2 L of 0.5 M solution is

(1) 35.6 mL (2) 42.1 mL

(3) 65.5 mL (4) 101.1 mL

84. Angular momentum of an electron in third stationary orbit of hydrogen atom will be

(1) $3\frac{h}{2}$ (2) $3h$

(3) $\frac{h}{3}$ (4) $\frac{2h}{3}$

85. Energy required to excite the electron in hydrogen atom from first orbit to second orbit is

(1) 1.6×10^{-18} J (2) 2.4×10^{-19} J

(3) 3.6×10^{-20} J (4) 3.6×10^{-17} J

SECTION-B

86. Number of degenerate orbitals present in the third shell of He^+ ion is

(1) 6 (2) 5

(3) 9 (4) 3

87. Consider the following statements about photoelectric effect

a. The number of electrons ejected is proportional to the frequency of light used.

b. There is no time lag between the striking of light beam and the ejection of electrons from metal surface.

c. The kinetic energy of ejected electrons increase with the increase of frequency of light used.

The correct statement(s) is/are

(1) (b) only (2) (a) and (b) only

(3) (a), (b) and (c) (4) (b) and (c) only

88. Match the atom/ion given in column I with the number of unpaired electrons given in column II.

	Column I		Column II
a.	N	(i)	5
b.	Cr	(ii)	4
c.	Mn^{2+}	(iii)	3
d.	Fe^{2+}	(iv)	6

The correct match is

(1) a(iii), b(ii), c(i), d(iv) (2) a(iii), b(iv), c(i), d(ii)

(3) a(iii), b(iv), c(ii), d(i) (4) a(ii), b(iv), c(i), d(iii)

89. The pair of compounds which does not follow Law of multiple proportions is

(1) CO_2 and CO (2) SO_2 and SO_3

(3) H_2S and H_2O (4) NO_2 and N_2O

90. The quantum number which gives information about the spatial orientation of orbital with respect to standard set of co-ordinate axis is

(1) Azimuthal quantum number

(2) Spin quantum number

(3) Principal quantum number

(4) Magnetic orbital quantum number

91. If a radiation of energy 5.25 eV falls on a metal surface, then the maximum kinetic energy of ejected electron will be (work function of metal = 3.25 eV)

(1) 3.2×10^{-19} J (2) 1.6×10^{-18} J

(3) 3.2×10^{-20} J (4) 1.6×10^{-17} J

92. If all the given molecules are moving with same velocity, then the molecule whose wavelength will be maximum is

(1) O_2 (2) CH_4

(3) He (4) N_2

93. Correct order of wavelength of the given radiations is

(1) UV rays > IR > Microwave > Radio waves

(2) Radio waves > Microwave > IR > UV rays

(3) Radio waves > IR > Microwaves > UV rays

(4) UV rays > Microwaves > IR > Radio waves

94. How many grams of concentrated nitric acid solution should be used to prepare 200 mL of 4.0 M HNO_3? (concentrated HNO_3 is 70% (w/w) HNO_3)

(1) 50.4 g

(2) 72 g

(3) 65 g

(4) 42.6 g

95. **Statement I:** Law of definite proportions states that a given compound always contains exactly the same proportion of elements by weight.

Statement II: Mass percentage of copper in $CuCO_3$ is 44.5%. (Atomic mass of Cu = 63.5 u)

In light of the above statements, choose the correct answer.

(1) Statement I is incorrect but statement II is correct

(2) Statement I is correct but statement II is incorrect

(3) Both statement I and statement II are correct

(4) Both statement I and statement II are incorrect

96. Mass of $CaCO_3$ required to react completely with 50 mL of 1.5 M HCl solution is

(1) 6.45 g (2) 10.25 g

(3) 5.1 g (4) 3.75 g

97. Consider the following statements about Dalton's atomic theory

a. Atoms of a given element have different mass.

b. Matter consists of indivisible atoms.

c. Atoms are neither created nor destroyed in a chemical reaction.

The correct statements are

(1) (a) and (b) only (2) (b) and (c) only

(3) (a) and (c) only (4) (a), (b) and (c)

98. If mass percentage of carbon in a hydrocarbon is 80%, then the empirical formula of the compound will be

(1) C_2H_3

(2) CH_2

(3) CH

(4) CH_3

99. Set of Isoelectronic species is

(1) N_2, O_2^+ and CO

(2) CN^-, N_2 and NO^+

(3) F_2, O_2^- and NO

(4) C_2, N_2^+ and B_2^-

100. The number of nitrogen atoms present in 20 litre of air containing 78% nitrogen by volume at STP is

(1) $2.5N_A$ (2) $4.1N_A$

(3) $6.5N_A$ (4) $1.4N_A$

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[BOTANY]

SECTION-A

101. Select the **incorrect** statement from the following.
(1) Reproduction either sexual or asexual is fundamental characteristic of living organisms
(2) Growth is not a defining feature of living beings because it is not found in some of the simplest organisms
(3) Both anabolism and catabolism occur in an organism
(4) All characteristic features of living beings are not necessary to be defining feature

102. The taxonomical aid that is based on couplets and leads is
(1) Herbarium (2) Manual
(3) Key (4) Monograph

103. Which of the following specific epithets is not associated with kingdom Plantae?
(1) *melongena* (2) *tigris*
(3) *indica* (4) *aestivum*

104. Select the **correct** option regarding the contribution of Robert Brown.
(1) He first saw and described the live cells
(2) He first classified the living and non-living
(3) He discovered the chromosomes in the nucleus
(4) He discovered the nucleus in the cell

105. Regarding peroxisome, find out the option that is **incorrect**.
(1) It is present in the cytoplasm
(2) It is a membrane bound structure
(3) It contains enzymes
(4) It is included in endomembrane system

106. Satellite is a structure in cell which is associated with
(1) Chromosome
(2) Nuclear membrane
(3) Endoplasmic reticulum
(4) Centrosome

107. Which of the following sets of structures are present in **both** prokaryotic and eukaryotic cells?
(1) Cilia, flagella and pili
(2) Ribosomes, cell wall and plasma membrane
(3) Chromosomes, nuclear membrane and inclusion bodies
(4) Ribosomes, lysosomes and vacuoles

108. The smallest living cell known lacks
(1) DNA as genetic material
(2) Cell wall
(3) Ribosomes with 50S subunits
(4) Proteins in plasma membrane

109. Regarding the procedure developed by Gram, bacteria can be classified into two groups. This classification is on the basis of
(1) Chemical nature of cytoplasmic inclusion bodies
(2) RNA types in ribosomes
(3) Presence and absence of proteins in the cell wall
(4) The differences in the cell envelopes and response to the particular stain

110. Match the following columns and choose the **correct** option

Column I	Column II
a. Cytoplasm	(i) Circular DNA
b. Centriole	(ii) Main arena of cellular activities
c. Plasmid	(iii) Contains chromatin
d. Nucleoplasm	(iv) Helps in cell division

(1) a(iv), b(i), c(ii), d(iii) (2) a(ii), b(i), c(iv), d(iii)
(3) a(iii), b(ii), c(iv), d(i) (4) a(ii), b(iv), c(i), d(iii)

111. The outermost layer in the three enveloped covering in bacteria can be all, **except**
(1) Mucopeptide cell wall
(2) Loose sheath of slime
(3) Glycocalyx
(4) Thick and tough capsule

112. Enzymes in lysosome are optimally active at
(1) Alkaline pH
(2) Neutral pH
(3) Acidic pH
(4) Basic pH

113. Filament, hook and basal body are the parts of
(1) A locomotary structure present in *Paramoecium*
(2) A structure present in all eukaryotic cells which is responsible for locomotion
(3) A double membrane bound structure present in animal cells only
(4) A filamentous extension responsible for locomotion in bacteria

114. The structures in some bacteria that help to attach the bacteria to the host tissue are
(1) Mesosome (2) Fimbriae
(3) Flagella (4) Cilia

115. Select the **incorrect** statement regarding the component in prokaryotic cell which is site of protein synthesis.
(1) It is a non-membrane bound structure
(2) It is found to be associated with plasma membrane
(3) It has two subunits-60S and 40S
(4) It forms polysome

116. Inclusion bodies in prokaryotes do not include
(1) Cyanophycean granules
(2) Glycogen granules
(3) Gas vacuoles
(4) Plasmids

117. Select the **incorrect** match from the following.
(1) Cytoskeletal structure – Present in animal cells
(2) 70S ribosomes – Absent in higher plants
(3) Large central vacuole – Absent in animal cells
(4) Organised nucleus – Has nuclear envelope

118. Identify the following statements as true (T) or false (F) and choose the option accordingly.
A. Cell membrane is mainly composed of sugars and lipids.
B. Polar heads of lipids are towards inner side of the plasma membrane.
C. Major lipids in cell membrane are phospholipids
D. Integral proteins are partially or totally buried in the membrane.

	A	B	C	D
(1)	T	F	T	F
(2)	F	F	T	T
(3)	F	T	T	F
(4)	T	F	F	T

119. The layer that glues the neighbouring plant cells together is (i) composed of (ii)
Select the **correct** option to fill in the blanks (i) and (ii)

	(i)	(ii)
(1)	Middle lamella	Calcium carbonate
(2)	Cell wall	Galactans
(3)	Middle lamella	Calcium pectate
(4)	Secondary wall	Hemicellulose

120. The endoplasmic reticulum that bears ribosomes on its surface is frequently observed in the cells which are
(1) Involved in the synthesis of glucose but not proteins
(2) Actively involved in protein synthesis and secretion
(3) Without nucleus such as mature RBC
(4) Not showing any biochemical processes

121. Match the following columns and choose the **correct** option.

Column I	Column II
a. Amyloplast	(i) Membrane of vacuole
b. Chloroplast	(ii) Stores starch
c. Tonoplast	(iii) Stores oils
d. Elaioplast	(iv) Traps sunlight energy

(1) a(iii), b(iv), c(ii), d(i) (2) a(iii), b(i), c(iv), d(ii)
(3) a(ii), b(iii), c(iv), d(i) (4) a(ii), b(iv), c(i), d(iii)

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(1) Alkaline pH
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(1) Mesosome (2) Fimbriae
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(1) It is a non-membrane bound structure
(2) It is found to be associated with plasma membrane
(3) It has two subunits-60S and 40S
(4) It forms polysome

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(1) Cyanophycean granules
(2) Glycogen granules
(3) Gas vacuoles
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C. Major lipids in cell membrane are phospholipids
D. Integral proteins are partially or totally buried in the membrane.

	A	B	C	D
(1)	T	F	T	F
(2)	F	F	T	T
(3)	F	T	T	F
(4)	T	F	F	T

119. The layer that glues the neighbouring plant cells together is __(i)__ composed of __(ii)__ .
Select the **correct** option to fill in the blanks (i) and (ii).

	(i)	(ii)
(1)	Middle lamella	Calcium carbonate
(2)	Cell wall	Galactans
(3)	Middle lamella	Calcium peclate
(4)	Secondary wall	Hemicellulose

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(1) Involved in the synthesis of glucose but not proteins
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Column I	Column II
a. Amyloplast	(i) Membrane of vacuole
b. Chloroplast	(ii) Stores starch
c. Tonoplast	(iii) Stores oils
d. Elaioplast	(iv) Traps sunlight energy

(1) a(iii), b(iv), c(ii), d(i) (2) a(iii), b(i), c(iv), d(ii)
(3) a(ii), b(iii), c(iv), d(i) (4) a(ii), b(iv), c(i), d(iii)

122. Select the **incorrect** statement regarding centrosome.
(1) It contains two cylindrical structures
(2) It is surrounded by amorphous pericentriolar material
(3) Its components have nine doublets of peripheral microtubules
(4) The cylindrical structure in it has an organization like the cartwheel

123. A cell cycle involves
a. DNA replication
b. Cell division
c. Cell growth
d. Nuclear division
The correct ones are
(1) (a), (b) and (d) only
(2) All (a), (b), (c) and (d)
(3) (a) and (d) only
(4) (a), (c) and (d) only

124. The most active stage of a cell cycle that occupies most of the time of the cell cycle is
(1) Interphase (2) Cytokinesis
(3) G_2 phase (4) Mitotic phase

125. The phase of a cell cycle in which cell does **not** proliferate but remains metabolically active is called
(1) G_0 phase (2) Synthesis phase
(3) Resting phase (4) Gap 1 phase

126. The amount of DNA doubles but the number of chromosomes remains the same. This statement is true for a cell present in
(1) G_2 phase (2) M phase
(3) G_1 phase (4) S phase

127. Which of the following functions occur in both G_1 phase and G_2 phase of cell cycle?
(1) Separation of chromatids and synthesis of RNA
(2) Duplication of cell organelles and protein synthesis
(3) DNA replication and centriole duplication
(4) Protein synthesis and centriole duplication

128. The most dramatic period of cell cycle involves
(1) Two stages of karyokinesis and a cytokinesis
(2) Four stages of nuclear division only
(3) Four stages of karyokinesis followed by cytokinesis
(4) One stage of cytokinesis followed by karyokinesis

129. Select the **odd** one out w.r.t. prophase of mitosis.
(1) Chromosome material condenses
(2) Centrosomes move towards opposite poles
(3) Nuclear envelope disintegrates
(4) Spindle fibres of mitotic apparatus dissolve

130. In which stage of mitotic division, spindle fibres attach to the kinetochores?
(1) Anaphase (2) Telophase
(3) Metaphase (4) Cytokinesis

131. The feature which is unique to the metaphase I of meiosis is
(1) Formation of metaphasic plate
(2) Separation of sister chromatids
(3) Alignment of bivalent chromosomes on the equatorial plate
(4) Movement of chromosomes towards equatorial plate

132. Arrange the following events of mitotic division in correct order and choose the option accordingly.
a. Condensation of chromosomes is completed
b. Chromosomes cluster at opposite spindle poles
c. Golgi complex and nucleolus disappear
d. Chromatids move to opposite poles
(1) d → c → a → b
(2) c → a → d → b
(3) b → a → c → d
(4) c → d → b → a

133. How many duplication(s) of centrosome occurs in one complete meiotic division in an animal cell?
(1) Two
(2) One
(3) Four
(4) Three

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134. Select the **incorrect** match from the following.

(1) Zygotene – Occurrence of synapsis

(2) Diakinesis – Disappearance of nucleolus

(3) Leptotene – Coiling of chromatin

(4) Pachytene – Chiasmata shifts to the tip of chromosome

135. In cell division, the enzyme recombinase is involved during the process that

(1) Splits centromere and separates sister chromatids apart

(2) Causes genetic variations in all the four daughter cells

(3) Align the homologous chromosomes together

(4) Is responsible for the formation of synaptonemal complex

SECTION-B

136. Identify the **incorrect** match from the following.

(1) *Allium* – Angiospermae

(2) *Petunia* – Monocytoledonae

(3) Wheat – Poaceae

(4) Leopard – Felidae

137. The set of rules and recommendations dealing with the formal names of an organism is given in ICBN. Which amongst the following characteristics would be there in this organism?

a. Presence of both membrane bound and non-membrane bound organelles.

b. Heterotrophic mode of nutrition is predominant.

c. Presence of rigid cell wall made up of cellulose.

d. Occurrence of 70S ribosomes in the cytosol.

The **correct** one(s) is/are

(1) All a, b, c and d (2) c only

(3) a, c and d only (4) a and c only

138. The fundamental structural and functional unit of all living organisms is

(1) DNA

(2) Protoplasm

(3) Cell

(4) Nucleus

139. Identify the following statements as true (T) or false (F) and choose the option accordingly.

A. Centromere is an essential part of every chromosome.

B. Telocentric chromosomes lack primary constriction.

C. Secondary constrictions are non-staining regions in the chromosomes.

	A	B	C
(1)	T	F	T
(2)	F	T	T
(3)	F	F	T
(4)	T	T	F

140. According to the cell theory given by Rudolf Virchow.

(1) Genetic information is stored in the chromosomes in a cell

(2) All cells arise from pre-existing cells

(3) Different cell organelles perform different types of activities

(4) Same type of cells together form a tissue

141. A special structure in prokaryotic cells that help in cell wall formation, DNA replication and distribution to daughter cells

(1) Does not have respiratory enzymes

(2) Is a non-membranous structure

(3) Is formed by invagination of plasma membrane

(4) Surrounds the region that has the genomic DNA

142. Disc shaped structure present on the centromere is known as

(1) Acrosome (2) Kinetochore

(3) Chromatin (4) Telomere

143. Polyribosome is an association of

(1) Many RNA molecules in a ribosome

(2) Many ribosomes attached to a single mRNA

(3) Several mRNA molecules around a single ribosome

(4) Different types of ribosomes within a single organelle

144. In a cell, all of the following are the locations where ATP is synthesized, **except**
(1) Nucleoplasm (2) Mitochondrion
(3) Chloroplast (4) Cytoplasm

145. The structure which is important for osmoregulation and excretion in *Amoeba* is
(1) Gas vacuole (2) Lysosome
(3) Contractile vacuole (4) Food vacuole

146. If a yeast cell completes a cell cycle in 90 minutes, what would be the approximate time for its M phase?
(1) 45 minutes (2) 85 minutes
(3) 4.5 minutes (4) 8.5 minutes

147. Which of the following is **not** the significance of meiosis?
(1) Increasing genetic variability in the population
(2) Formation of gametes
(3) Production of new cells for healing and regeneration
(4) Maintenance of chromosome number in sexually reproducing organisms

148. During which stage of meiosis, chiasmata becomes visible?
(1) Anaphase I
(2) Diplotene
(3) Telophase I
(4) Diakinesis

149. The event that occurs during anaphase I of meiosis is
(1) Separation of homologous chromosome from each other
(2) Splitting of centromere
(3) Formation of nuclear membrane
(4) Disappearance of nucleolus

150. Phragmoplast is formed by Golgi complex and grows centrifugally to form cell plate. This event regarding cytokinesis occurs in
(1) All the organisms that have cell wall
(2) Plant cells
(3) Bacteria and unicellular eukaryotes
(4) All eukaryotes

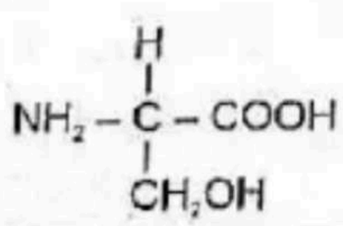
[ZOOLOGY]

SECTION-A

151. Select the cells present in areolar connective tissue that engulf and digest microbes, foreign particles and degenerative cells.
(1) Fibroblasts (2) Neuroglial cells
(3) Mast cells (4) Macrophages

152. On being activated, which of the following cells of connective tissue produce and secrete fibres and histamine, respectively?
(1) Fibroblasts and adipocytes
(2) Fibroblasts and mast cells
(3) Macrophages and fibroblasts
(4) Mast cells and macrophages

153. Choose the correct option w.r.t. name of amino acid given below :



(1) Glycine
(2) Serine
(3) Alanine
(4) Glutamic acid

154. Which cells of neural tissue perform many house keeping functions and are non-excitable?
(1) Dendrites
(2) Neuroglial cells
(3) Macrophages
(4) Osteocytes

155. White fibres which provide tensile strength against mechanical stresses and abundantly reported in connective tissue as intercellular ground substance, are
(1) Inulin
(2) Collagen
(3) Cellulose
(4) Myofibrils

156. Complete the analogy by selecting the correct option.

Smooth muscle fibres : Fusiform :: Cardiac muscle fibres : _____.

(1) Voluntary

(2) Skeletal

(3) Cylindrical

(4) Non-striated

157. Read the following statements carefully.

Statement I: Tendons are inelastic cords which connect skeletal muscles to bones.

Statement II: Fibres and fibroblasts are compactly packed in the dense connective tissue.

Statement III: Dense irregular connective tissue is present in the skin.

Choose the correct option.

(1) Statements I, II and III are correct.

(2) Statements I and II are correct but statement III is incorrect.

(3) Statements I and II are incorrect but statement III is correct.

(4) Only statements I and III are correct.

158. The cells of cartilage are enclosed in small cavities within the matrix secreted by _____.

Select the correct option that fills the blank.

(1) Chondrocytes

(2) Osteocytes

(3) Fibroblasts

(4) Surrounding epithelium

159. Bones of limb in adult humans are chiefly formed by ossification of

(1) Tendons (2) Ligaments

(3) Cartilage (4) Areolar tissue

160. On the basis of chemical analysis of living tissue of animals, which of the following is the most abundant protein in them?

(1) RuBisCO

(2) Chitin

(3) Collagen

(4) Trypsin

161. Ribose is differentiated from deoxyribose in having

(1) Two extra oxygen at 2nd carbon

(2) No oxygen at 3rd carbon

(3) No oxygen at carbon number 2

(4) Hydroxyl group at 2nd carbon

162. Read the statements carefully and select the incorrect one.

(1) When a neuron is suitably stimulated, an electrical disturbance is generated which travels along its plasma membrane.

(2) Ligaments attach one bone to another.

(3) Compound epithelium covers the moist surface of buccal cavity.

(4) The products of exocrine glands are directly secreted into the fluid bathing the gland.

163. Which of the following statements about enzymes is wrong?

(1) Enzymes act best at their optimum temperature and pH.

(2) All the cells of the organism contain the same set of enzymes.

(3) Enzymes are biochemical catalysts available within the cells.

(4) Enzymes lower the barrier of activation energy of spontaneous reactions in the cell.

164. In case of lecithin, choline is attached with

(1) Third hydroxyl group of trihydroxy propane without phosphoric acid.

(2) Phosphate group joined to one of the –OH group of trihydroxy propane

(3) Second hydroxyl group of glycerol with phosphate

(4) First hydroxyl group of glycerol with phosphate.

165. Choose the **correct** option w.r.t. location and function of glandular epithelium.

	Location	Function
(1)	Lung's alveoli	Attach skeletal muscle to bones
(2)	Intestine	Secretion
(3)	Heart	Contraction
(4)	Blood vessels	Secretion and absorption

166. Match the type of animal tissue in column I with the location given in column II.

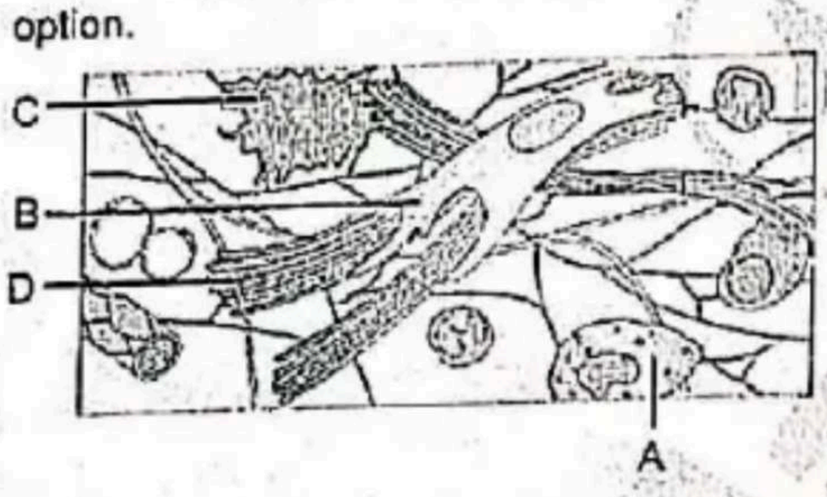
Column I	Column II
a. Simple columnar epithelium	(i) Wall of heart
b. Cardiac muscles	(ii) Tip of nose
c. Adipose tissue	(iii) Inner lining of stomach and intestine
d. Cartilage	(iv) Beneath the skin
	(v) Blood vessels

Choose the correct option.

(1) a(i), b(ii), c(iv), d(iii) (2) a(iii), b(v), c(ii), d(i)

(3) a(iii), b(i), c(iv), d(ii) (4) a(ii), b(iii), c(i), d(iv)

167. In given diagram of certain type of connective tissue, identify A, B, C and D and select the correct option.



	A	B	C	D
(1)	Fibroblast	Macrophage	Collagen fibres	Mast cell
(2)	Mast cell	Fibroblast	Macrophage	Collagen fibres
(3)	Macrophage	Collagen fibres	Mast cells	Fibroblast
(4)	Collagen fibres	Mast cell	Fibroblast	Macrophage

168. Which connective tissue is characterised by the presence of collagen fibres, macrophage, fibroblasts and mast cells?

(1) Blood
(2) Cartilage
(3) Smooth muscles
(4) Areolar tissue

169. Wall of blood vessels, stomach and intestine are made of A muscles, which are B in nature.

Fill the blanks A and B respectively with suitable terms to make the statement correct.

(1) Skeletal, voluntary
(2) Skeletal, involuntary
(3) Smooth, involuntary
(4) Striped, involuntary

170. **Assertion (A):** Neurons make up more than one half the volume of neural tissue in our body.

Reason (R): The neuroglial cells of the neural system protect and support neurons.

In the light of above statements, choose the correct option.

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A).
(2) Both (A) and (R) are correct and (R) is not the correct explanation of (A).
(3) (A) is correct but (R) is incorrect.
(4) (A) is incorrect but (R) is correct.

171. Read the following statements and choose the option with incorrect statements only.

a. In tissue, a group of similar cells along with intercellular substances perform a specific function.
b. Organ systems split up the work in a way that exhibits division of labour.
c. In epithelium, cells are loosely packed with little or no intracellular materials.
d. Cell junctions hold smooth muscle fibres together and they are bundled together in a epithelial tissue sheath.

(1) a and b (2) b and c
(3) a, b and c (4) c and d

172. Read the following and choose the incorrect one w.r.t. acid soluble pool.

(1) Represent roughly the cytoplasmic components
(2) Lipids are found in it
(3) Molecules ranging from 18 to 800 Da
(4) Contains most of the micromolecules

@DefeatNEET

173. Select the **incorrect** statement w.r.t. nucleic acids (DNA).

(1) It exhibit a wide variety of secondary structures.

(2) One of the secondary structure exhibited by DNA is the famous Watson-Crick model.

(3) The two strands of polynucleosides are antiparallel.

(4) The backbone is formed by the sugar phosphate-sugar chain.

174. How many of the given biomolecules in box are secondary metabolites?

Concanavalin-A, GLUT-4, Collagen, Cellulose, Vinblastin, Scents, Codeine, Ricin, Trypsin, Insulin

Choose the **correct** option.

(1) 7

(2) 4

(3) 6

(4) 5

175. **Assertion (A):** Dietary proteins are the source of essential amino acids.

Reason (R): A protein is a heteropolymer having about 20 types of amino acids, which can be essential or non-essential.

In the light of above statements, choose the correct option.

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A).

(2) Both (A) and (R) are correct but (R) is not correct explanation of (A).

(3) (A) is correct but (R) is incorrect.

(4) (A) is incorrect but (R) is correct.

176. Choose the **correct** statement.

(1) Lipids are polymeric micromolecules.

(2) Adenine is a substituted pyrimidine.

(3) Physicists conjure up the molecular structures of proteins at four levels.

(4) Tertiary structure is absolutely necessary for the many biological activities of proteins.

177. Formation of both peptide bonds and glycosidic bonds involves

(1) Hydration

(2) Esterification

(3) Dehydration

(4) Acidification

178. Read the statements carefully and choose the **incorrect** one.

(1) Right end of polysaccharide is called reducing end while left end is called non-reducing end.

(2) Starch can hold I_2 molecules in its helical secondary structures but cellulose lack complex helices and cannot hold iodine.

(3) Starch and glycogen are branched polysaccharides.

(4) Starch in plants and glycogen in animals are store house of essential amino acids.

179. Choose the **correct** match.

	Nature		Amino acid
(1)	Acidic	–	Valine
(2)	Basic	–	Lysine
(3)	Neutral	–	Glutamic acid
(4)	Aromatic	–	Glycine

180. Which of the following statements is **incorrect**?

(1) Some secondary metabolites have ecological importance.

(2) Primary metabolites have identifiable functions.

(3) Secondary metabolites like spices and pigments are useful to human welfare.

(4) Secondary metabolites are not found in microbes and fungi etc.

181. How many 'C' atoms are reported in each molecule of arachidonic acid and palmitic acid respectively?

(1) 16, 16

(2) 20, 16

(3) 16, 18

(4) 20, 18

182. Each nucleotide is made up of

(1) Nitrogen containing carbohydrate and pentose sugar with phosphate

(2) Cyclic nitrogenous base + Pentose sugar and one to three phosphate group

(3) (Nitrogen base + Hexose sugar + phosphate)$_n$

(4) (Amino sugar + Ribose + phosphate)$_n$

183. NAD and NADP contains the vitamin

(1) Ascorbic acid

(2) Niacin

(3) Thiamine

(4) Riboflavin

184. A crevice or pocket of an enzyme into which the substrate fits is a/an

(1) Allosteric site

(2) Regulatory site

(3) Active site

(4) Restriction site

185. Read the statements carefully and choose the correct one.

(1) Assembly of a protein from amino acids does not require energy input.

(2) Catabolic pathways consumes energy.

(3) The metabolic pathway from glucose to lactic acid occurs in 10 anabolic steps called glycolysis.

(4) As and when needed, bond energy (ATP) is utilised for biosynthetic, osmotic and mechanical work that we perform.

SECTION-B

186. Which of the following junctions facilitate the cells to communicate with each other by connecting the cytoplasm of adjoining cells for rapid transfer of ions, small molecules and sometimes big molecules?

(1) Adhering junctions

(2) Gap junctions

(3) Tight junctions

(4) Hemidesmosomes

187. Match column I and column II and choose the correct option w.r.t. muscle fibres.

Column I	Column II
a. Skeletal	(i) Uninucleated
b. Smooth	(ii) Multinucleated
c. Cardiac	(iii) Uninucleated and branched

(1) a(ii), b(iii), c(i)

(2) a(ii), b(i), c(iii)

(3) a(iii), b(i), c(ii)

(4) a(i), b(ii), c(iii)

188. Proteinaceous enzymes exhibit substrate specificity. If an enzyme 'X' can act on different substrates, its K_m value will

(1) Remain same for different substrates

(2) Differ from substrate to substrate

(3) Remain same even in presence of competitive inhibitior

(4) Change with the type of substrate but V_{max} is never achieved

189. A healthy person in his twenties got injuries resulting in damage and death of several tissues and cells. Which of the cells are likely to be severely damaged, in case of reduced control over the body's responsiveness?

(1) Chondrocytes

(2) Osteocytes

(3) Mast cells

(4) Neurons

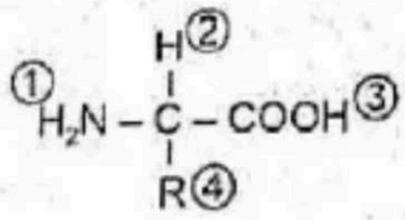
190. Match column I and column II w.r.t. structure and their location.

Column I	Column II
a. Cuboidal epithelium	(i) Cardiac muscle fibres
b. Ciliated epithelium	(ii) Inner surface of fallopian tubes
c. Columnar epithelium	(iii) Lung's alveoli
d. Squamous epithelium	(iv) Lining of ducts of glands
e. Intercalated disc	(v) Nuclei are located at the base

(1) a(iv), b(ii), c(v), d(iii), e(i)
(2) a(i), b(ii), c(iii), d(iv), e(v)
(3) a(ii), b(iii), c(iv), d(v), e(i)
(4) a(iii), b(ii), c(iv), d(i), e(v)

191. % weight of oxygen, sulphur, magnesium and nitrogen in human body respectively is
(1) 0.3, 65.0, 3.3 and 0.1
(2) 65.0, 0.3, 0.1 and 3.3
(3) 65.0, 3.3, 46.6 and 0.1
(4) 3.3, 0.1, 0.3 and 65.0

192. Which of the two groups of following formula are involved in peptide bond formation between successive amino acids?



(1) 2 and 3
(2) 3 and 1
(3) 1 and 4
(4) 4 and 3

193. Read the following features
a. Spindle shaped muscle fibres
b. Uninucleated and involuntary
c. They are abundantly reported in iris of the eye
Which type of muscular tissue exhibit given features?
(1) Skeletal
(2) Smooth
(3) Striped
(4) Cardiac

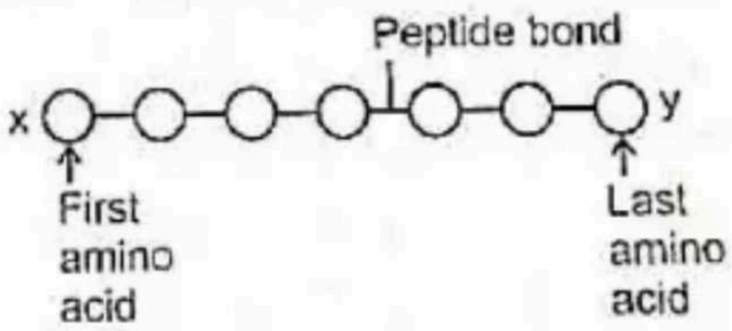
194. Choose the option representing only homopolymeric biomolecules.
(1) Starch, Cellulose, Glycogen and Inulin
(2) Insulin, Cellulose, Chitin, Collagen
(3) Polypeptide, Inulin, Lecithin, Cellulose
(4) Lipids, Cholesterol, Chitin, Inulin

195. Match column I and column II carefully.

Column I	Column II
a. Pigments	(i) Concanavalin-A
b. Terpenoids	(ii) Monoterpenes
c. Alkaloids	(iii) Morphine
d. Lectins	(iv) Carotenoids
e. Toxins	(v) Abrin

Choose the correct option.
(1) a(i), b(ii), c(iii), d(iv), e(v)
(2) a(iv), b(ii), c(iii), d(i), e(v)
(3) a(ii), b(iii), c(iv), d(v), e(i)
(4) a(iii), b(ii), c(i), d(v), e(iv)

196. Choose the **correct** statement w.r.t. given figure.



(1) 'x' is 'C' terminal amino acid and 'y' is N-terminal amino acid.
(2) Both 'x' and 'y' are N-terminal amino acids
(3) Both 'x' and 'y' are C-terminal amino acids
(4) 'x' is N-terminal amino acid and 'y' is C-terminal amino acid

197. What will be the molecular formula of a polypeptide having 10 glycine in a series?

(1) $C_{10}H_{50}O_{20}N_{10}$

(2) $C_{20}H_{32}O_{11}N_{10}$

(3) $C_{10}H_{48}O_{19}N_{10}$

(4) $C_{19}H_{48}O_{19}N_{10}$

198. Proteins carry out many functions in living organisms. The enormous diversity of protein molecules is mainly due to the diversity of

(1) Amino acids based on amino group only

(2) Peptide bonds formed

(3) Nature of R group and sequence of amino acids within the protein molecule

(4) Amino acids at C-terminal and N-terminal in protein molecule

199. What happens when the succinate dehydrogenase enzymes action is inhibited by malonate?

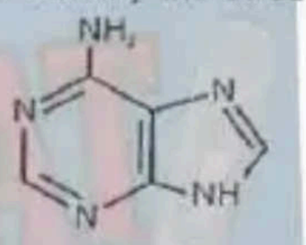
(1) Increase in V_{max} without affecting K_m

(2) Increase in K_m without affecting V_{max}

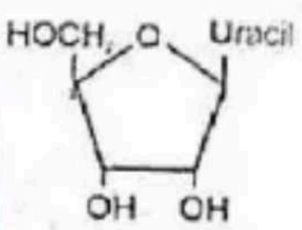
(3) Decrease in K_m without affecting V_{max}

(4) Decrease in V_{max} without affecting K_m

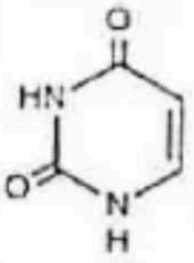
200. Identify the structures A, B and C given below.



A



B



C

Choose the correct option.

	A	B	C
(1)	Adenine	Uridine	Uracil
(2)	Guanine	Uracil	Adenine
(3)	Uracil	Adenine	Adenylic acid
(4)	Adenosine	Adenylic acid	Uracil

❑ ❑ ❑

For Answers & Solutions

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